10 September 1970
SUPERSEDING
MIL-C-35041A
8 February 1965

MILITARY SPECIFICATION

CHOCOLATE FLAVORED SIRUP; BEVERAGE, AND TOPPING (COLD)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers stabilized chocolate flavored sirup for use by the Armed Forces in the preparation of chocolate milk beverage or as a cold topping. The product is intended for use as an item of general issue.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-C-29 - Canned Subsistence Items, Packaging and Packing Of.

MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items.
MIL-L-35078 - Loads, Unit; Preparation of Nonperishable Subsistence In.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 8925

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MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-668 - Sanitary Standards for Food Plants.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Officer, Washington, D.C. 20402.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS

Official Methods of Analysis of the Association of Official Agricultural Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, Box 540, Benjamin Franklin Station, Washington, D.C. 20044.)

3. REQUIREMENTS

3.1 Materials.- All of the materials shall be clean, sound, wholesome, and free from any foreign odors, flavors, extraneous materials or insect infestation.

* 3.1.1 Chocolate liquor.- Chocolate liquor shall be manufactured from domestically roasted cocoa beans and shall conform to the Definitions and Standards for Foods of the Federal Food, Drug, and Cosmetic Act for cacao products. The chocolate liquor shall not be alkali processed.

* 3.1.2 Cocoa.- Cocoa shall be the medium fat or breakfast-type, made from domestically roasted cocoa beans and shall conform to the Definitions and Standards for Foods of the Federal Food, Drug, and Cosmetic Act for cacao products. The cocoa shall not be alkali processed.

3.1.3 Flavoring.- Flavoring shall be vanilla extract or imitation vanilla, or a combination of both. Spices may be used if desired.

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3.1.4 Salt.- Salt shall be white, refined sodium chloride, with or without anticaking agents. Iodized salt shall not be used.

* 3.1.5 Sweetening agents.- Sweetening agents shall be white, refined, cane or beet sugar, or liquid sugar, or corn sirup, dextrose, or combination thereof.

3.1.6 Stabilizer.- A stabilizer, if used, shall produce a product that meets the requirements of paragraph 3.2.1.

* 3.2 Finished product.- The finished product shall be sufficiently heat processed to reach 220°F. for 5 minutes. It shall be free from lumps and crystals. One part of the sirup by volume, when mixed with 11 parts of skim or whole milk by volume (either fresh or reconstituted), shall produce a palatable chocolate milk beverage having a distinct chocolate flavor and aroma.

* 3.2.1 The product shall contain not less than 66 percent by weight of solids derived from added sweetening agents. It shall contain not less than 49 percent sucrose. It shall contain not more than 11.0 percent of reducing sugar and not more than 9.5 percent nonreducing polysaccharides. It shall contain not less than 12 percent by weight of chocolate or cocoa solids (other than fat) or both. It shall be salted and flavored to the extent necessary to produce a palatable chocolate milk beverage or chocolate topping. It shall contain a suitable stabilizing agent or be processed in such a manner as to insure nonsettling of the ingredients for not less than 1 hour in milk beverages, when mixed in accordance with 3.2 and held at a temperature of 50°F.

* 3.3 All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

3.4 Workmanship.- The product shall be processed in establishments meeting the requirements of MIL-STD-668. It shall be prepared and processed with minimum delay between the various stages in production.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.2 Preacceptance inspection.- Product shall be rejected if produced in plants not meeting the requirements of MIL-STD-668.

* 4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.3.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.3.1.1 Ingredients.- Conformance of ingredients to identity requirements (kind) shall be ascertained by examination of labels, invoices or other valid documents. In addition, each ingredient shall be examined organoleptically, as necessary, to determine conformance to condition requirements of 3.1. The sample unit shall be one pound of ingredient from a primary container. The sample size shall be five. In the event a lot consists of less than five containers, a sample unit shall be extracted from each container in the lot. Nonconformance with identity requirements, or one or more sample units not conforming to condition requirements shall indicate an unacceptable ingredient and use of such shall be cause for rejection of the involved finished product.

* 4.3.2 Process examination.- Examination shall be performed during processing to determine if product is being heat processed as specified in 3.2. Time and temperature records shall be maintained. Nonconformance shall be cause for rejection of the involved finished product.

* 4.3.3 Examination of end item.- Examinations shall be in accordance with tables II and III and 4.3.3.1 through 4.3.3.4. Sampling plans for the aforementioned tables shall be in accordance with table I. The acceptable quality level (AQLs) shall be expressed as percent defective for table II and defects per hundred units for table III.

TABLE I.- Sampling plans

Table	Inspection level	Sample unit	Lot size expressed in	AQLs Major	AQLs Minor
II	S-3	1-filled can	cans	-	2.5
III	S-2	1-filled can	can	1.5	-

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TABLE II.- Examination for net weight 1/

Category	Defect
Minor	
201	Less than 8 pounds 1-1/2 ounce.

1/ Report results to nearest 1/4 ounce. Lot shall be rejected if sample data indicate a lot average net weight of less than 8 pounds 4 ounces.

TABLE III.- Examination of product 1/

Category	Defect
Major	
151	Presence of lumps or crystals in sirup.
152	Prepared beverage 2/ does not have a distinct chocolate flavor and aroma or not palatable.
153	Sirup solids settle within 1 hour when beverage is held at 50°F.

1/ Presence of foreign material (e.g., filth, wood, metal, glass, paint, insects, etc.) shall be cause for rejection of the lot.

2/ Prepare beverage by adding 1 fluid ounce of sirup to 11 fluid ounces of fresh or skim milk and mixing well. Make one determination per sample unit.

* 4.3.3.1 Examination of cans.- Examination of cans for external conditions, interior coating, and seam leakage shall be in accordance with the quality assurance provisions of PPP-C-29.

* 4.3.3.2 Examination of can label.- Examination shall be in accordance with the applicable quality assurance provisions of MIL-L-1497.

* 4.3.3.3 Examination of packing and shipping container.- Examinations shall be in accordance with the applicable quality assurance provisions of PPP-C-29.

* 4.3.3.4 Examination of unit loads.- Unit loads shall be examined in accordance with the quality assurance provisions of MIL-L-35078.

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* 4.3.4 Testing of finished product.- The sirup shall be tested for total added sweetening solids, sucrose, reducing sugars, nonreducing polysaccharides, and nonfat chocolate or cocoa solids. Test procedures shall be in accordance with 4.5. The sample for testing shall be a 1-quart composite derived from the number of cans indicated by inspection level S-2. Lot size shall be expressed in cans. Each test result shall be reported to the same unit or decimal place as specified for the pertinent requirement. Lot shall be rejected if one or more test results indicate nonconformance with requirements.

* 4.4 Standby test samples.- The Government reserves the right to withdraw and hold standby test samples of components or finished products or both (quantity of which shall be not more than twice that required by this specification) for inspection purposes. Samples not used will be returned to the supplier.

* 4.5 Test methods.- Examination and test procedures which differ from those specified herein, unless otherwise excepted, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government inspection activity determines that such procedures and controls do not provide, as a minimum such quality assurance, the supplier will use the procedures set forth herein. In case of dispute as to examination or test results, the procedures specified herein will govern.

* 4.5.1 Test procedures.-

* 4.5.1.1 Chemical analysis.- Chemical analysis for chocolate or cocoa shall be made in accordance with the following methods from Official Methods of Analysis of the Association of Official Agricultural Chemists; Chapter: Cacao Bean and Its Products, Section: Cacao Products Processed with Alkaline, Method: Chocolate Liquor.

* 4.5.1.2 Chemical analysis.- Chemical analysis for sucrose, reducing sugar and nonreducing polysaccharides of corn sirup on chocolate sirup shall be made in accordance with following methods from Official Method of Analysis of the Association of Official Agricultural Chemists; Chapter: Sugar and sugar products; section: Sugars, sirups and molasses; Method: Sucrose - Chemical Method, modified as follows.

* 4.5.1.2.1 Reagents.-

1. Invertase enzyme, purified powder-source Candida utilis (100 units/mg.) Code No. 32-008 (Miles Research Products Division, Elkhart, Indiana). Weigh 25 mg. into 250-ml volume flask and make to volume. Use 1 ml of enzyme for each ml of clarified filtrate.

INVERTASE UNIT: The amount of enzyme activity required to liberate 1 mg. of hexose when 1 ml of enzyme solution is mixed with 1 mg. of substrate and incubated for 3 minutes at 20°C.

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2. Gluco-amylase enzyme (30 units/mg.) Liquid Diazyme L30. (Miles Marschall Division, Elkhart, Indiana). Transfer 2 ml into 250-ml volume flask, mix and make to volume. Use 0.5 ml of enzyme for each ml of clarified filtrate. Refrigerated shelf life is not longer than six months after purchase.

GLUCO-AMYLASE UNITS: The amount of enzyme that will catalyze under assay conditions the production of one gram of dextrose in one hour.

3. Methyl Red Indicator, 0.1% in alcohol.

4. Lead acetate, $Pb(C_2H_3O_3)_2 \cdot 3H_2O$, Saturated soln.

5. Potassium oxalate, neutral.

6. Hydrochloric acid, 4N.

4.5.1.2.2 Procedure.-

1. Weigh 10g sample into 1000-ml volume flask, add ca 800 ml H_2O, and shake sample on mechanical shaker for ca 30 minutes.

2. Add 10 ml of lead acetate, mix, make to volume, and filter ca 200 ml volume through Whatman 2V filter paper into 250-ml Erelenmeyer flask.

3. Add slight excess of potassium oxalate (ca 1.5g) to filtered soln, shake, transfer to 250-ml centrifuge bottle, centrifuge for ca 5 min. at 1500 rpm, and decant through Whatman 2V filter paper into 250-ml stoppered Erlenmeyer flask,

4. Pipette 50 ml of the clarified filtrate (Step 3) into 400-ml beaker for determination of reducing sugar (AOAC, Chapter: Sugar and sugar products; section: Sugars, sirups and molasses; Method: Sucrose - Chemical Method).

5. For inversion of sucrose, pipette 20 ml of clarified filtrate, 20 ml of invertase, and 10 ml of H_2O into 400-ml beaker, add 1 drop of methyl red and ca 3 drops of 4N HCl to adjust to pH 4-5. Cover beaker with watch glass and incubate in mechanical convection oven for 0.5-1.0 hours at 55-60°C. Determine reducing sugar as in AOAC. (see 4 above)

6. For determination of total carbohydrate (sucrose, reducing sugar, and nonreducing polysaccharides), pipette 20 ml of clarified filtrate, 20 ml of invertase, and 10 ml of gluco-amylase into 400-ml beaker. Add 1 drop of methyl red, ca 3 drops 4N HCl to adjust to pH 4-5,

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and follow the procedure above for incubation and determination of reducing sugar as in AOAC. (see 4 above)

7. Fritted disc crucible, fine or medium porosity, may be used instead of crucible with asbestos mat.

8. Calculate as in AOAC. (see 4 above)

a. % reducing sugar before inversion =A
b. % sucrose = B
c. % total carbohydrate = C
d. % nonreducing polysaccharides = C-(A+B)

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Eight pounds, four ounces of chocolate flavored sirup shall be packaged in a size 603 by 700 can in accordance with the level A or C requirements of PPP-C-29, as specified (see 6.1).

5.2 Packing.- Packing shall be in accordance with level A, B or C requirements of PPP-C-29 as specified (see 6.1).

5.3 Labeling and marking.-

5.3.1 Cans.- Cans shall be labeled in accordance with MIL-L-1497. In addition, the label shall include the following information:

DIRECTIONS FOR USE

Chocolate flavored sirup may be used for making chocolate-flavored milk, or for topping on ice cream, pudding or cake.

For milk drinks, mix 1 part sirup by volume with 11 parts liquid milk by volume (1-1/2 tablespoons sirup to 8 fl. oz. milk).

5.3.2 Shipping containers.- Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Levels of packaging and packing required (see 5.1 and 5.2).

* 6.2 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only

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and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, MS
Air Force - 03

DSA-SS

Preparing activity:

Army - GL

Project No. 8925-0046

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-35041B CHOCOLATE FLAVORED SIRUP; BEVERAGE, AND TOPPING (COLD)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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